

Student Mental Health Early Warning System



Software Engineering Project Documentation

Submitted to:

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MILESTONE 1: Requirements

Problem Statement

University students are increasingly vulnerable to mental health issues such as anxiety, depression, and burnout. These conditions often go undetected until they escalate into crises. The Student Mental Health Early Warning System (SMHEWS) aims to proactively identify at-risk students using behavioral indicators, academic performance data, and self-reported assessments — enabling timely intervention by counselors and academic staff.

Project Scope

In Scope

- ✔ Student registration and secure login
- ✔ Mental health self-assessment questionnaires
- ✔ Behavioral and academic data monitoring
- ✔ Automated risk-level classification (Low / Medium / High)
- ✔ Counselor alert and notification system
- ✔ Anonymous reporting feature
- ✔ Progress tracking and wellness reports
- ✔ Resource library (articles, helplines, coping strategies)

Out of Scope

- ✗ AI-based diagnosis or medical prescriptions
- ✗ Video/audio therapy sessions
- ✗ Integration with hospital management systems

Functional Requirements

ID	Requirement Description
FR1	Student can register and login securely
FR2	Student can complete mental health self-assessment (PHQ-9 / GAD-7 based)
FR3	System calculates and displays a risk score (Low / Medium / High)
FR4	System sends automated alerts to counselors when High risk is detected
FR5	Counselor can view student risk profiles and history
FR6	Student can submit anonymous concern reports

ID	Requirement Description
FR7	System tracks mood trends over time with visual charts
FR8	System sends daily wellness check-in reminders
FR9	Student can access a mental health resource library
FR10	Admin can generate aggregate campus-wide wellness reports

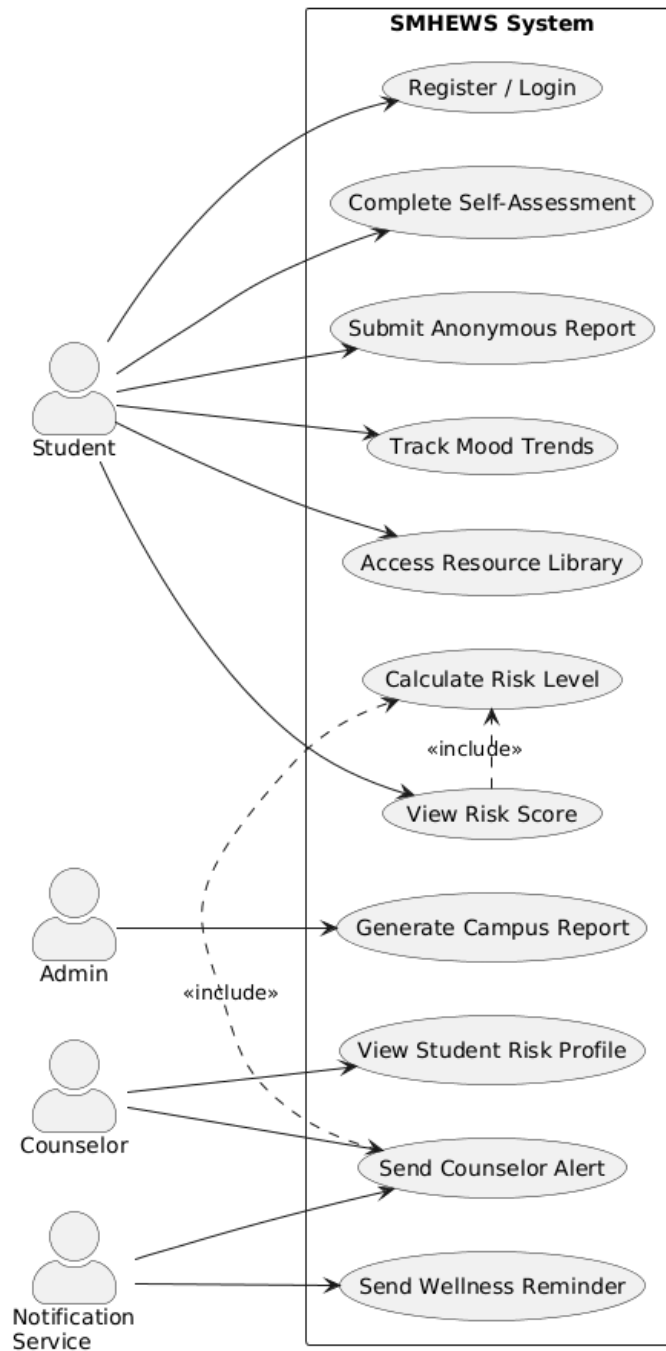
Non-Functional Requirements

ID	Requirement
NFR1	System must load within 3 seconds on standard network
NFR2	All student data must be encrypted at rest and in transit (AES-256 / TLS)
NFR3	Application must be accessible on Android, iOS, and web browsers
NFR4	UI must comply with WCAG 2.1 accessibility guidelines
NFR5	System must maintain 99.5% uptime
NFR6	Counselor alerts must be delivered within 60 seconds of risk detection

1. Use Case Diagram

The following Use Case Diagram illustrates all actors and their interactions with the SMHEWS system.

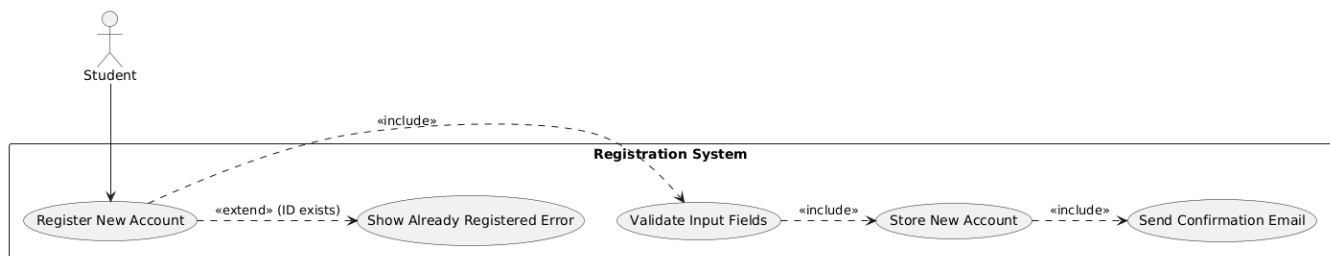
Use Case Diagram



2. Use Case Descriptions

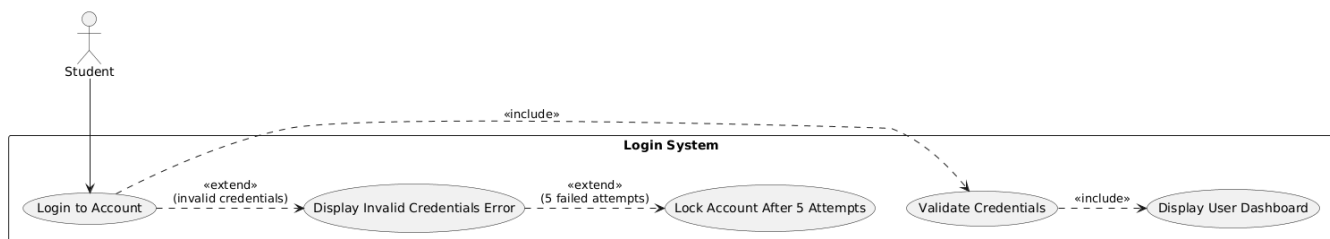
Use Case 1: Register

Actor	Student
Precondition	Student is not already registered.
Postcondition	A new student account is created and stored securely.
Main Flow	1. Student opens registration page. 2. Student enters name, student ID, email, and password. 3. System validates all input fields. 4. System creates the student account. 5. Confirmation email is sent.
Exception Flow	If student ID already exists, system shows 'Account already registered' error.



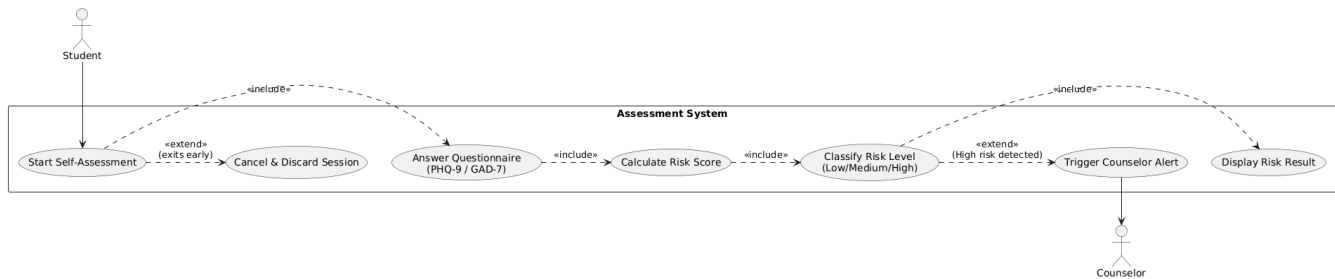
Use Case 2: Login

Actor	Student / Counselor / Admin
Precondition	User must already be registered.
Postcondition	User is authenticated and redirected to their dashboard.
Main Flow	1. User enters email and password. 2. System validates credentials against database. 3. Session token is generated. 4. User is redirected to dashboard.
Exception Flow	Invalid credentials: system shows error message and locks account after 5 failed attempts.



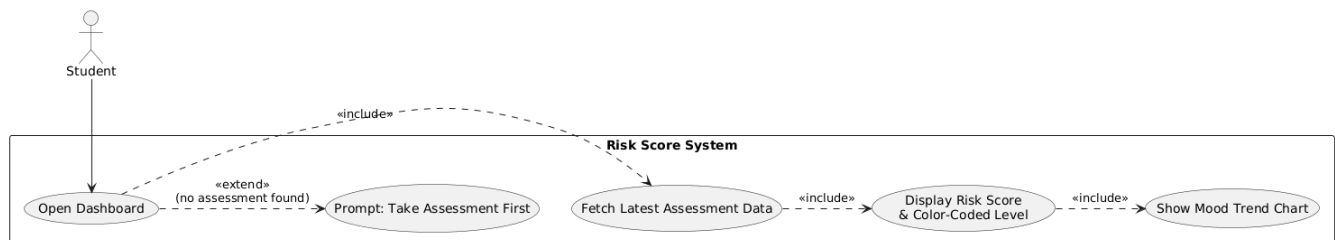
Use Case 3: Complete Self-Assessment

Actor	Student
Precondition	Student is logged in.
Postcondition	Assessment stored; risk score calculated and updated.
Main Flow	1. Student opens Assessment section. 2. Student answers PHQ-9 / GAD-7 questionnaire. 3. System scores the responses. 4. System classifies risk level (Low / Medium / High). 5. Result is displayed to the student. 6. If High risk, counselor alert is triggered.
Exception Flow	If student exits before completion, responses are not saved and score is not updated.



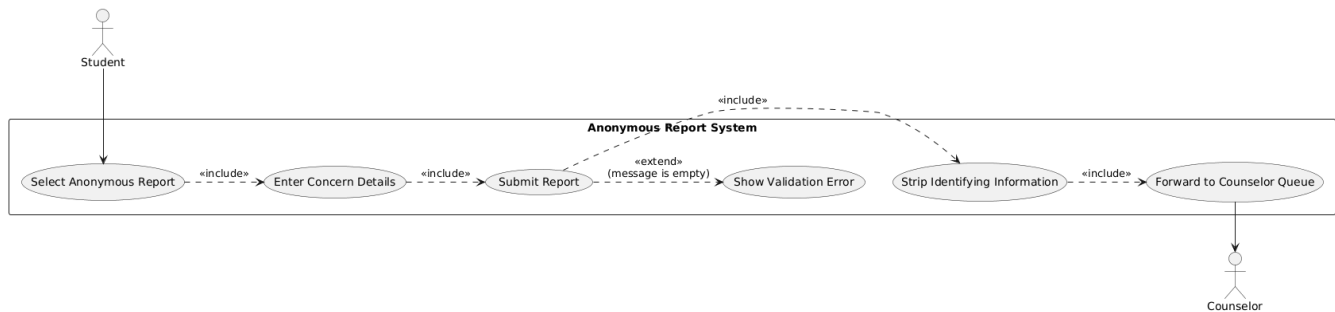
Use Case 4: View Risk Score

Actor	Student
Precondition	Student has completed at least one self-assessment.
Postcondition	Risk score and trend chart displayed.
Main Flow	1. Student opens Dashboard. 2. System fetches latest assessment data. 3. System displays current risk score and color-coded level. 4. Trend chart shows history over time.
Exception Flow	If no assessment has been completed, system prompts student to take assessment.



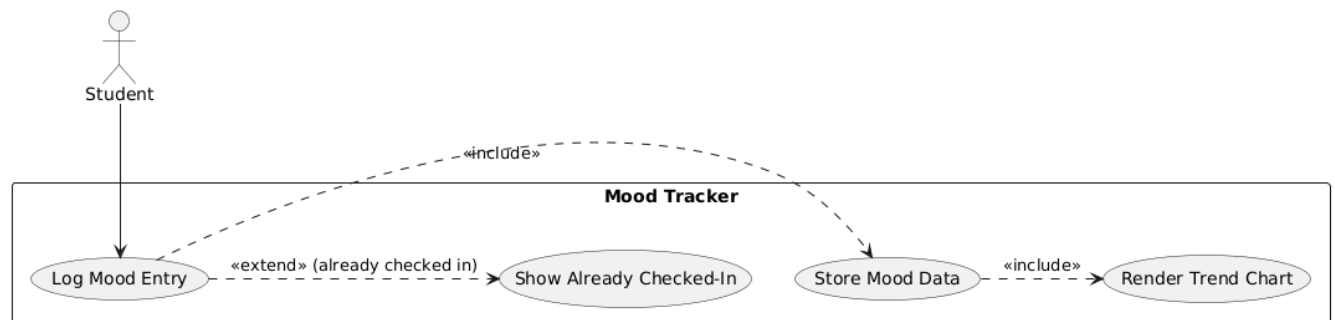
Use Case 5: Submit Anonymous Report

Actor	Student
Precondition	Student is logged in.
Postcondition	Anonymous concern report stored and forwarded to counselor.
Main Flow	1. Student selects Anonymous Report. 2. Student types concern details. 3. Student submits the form. 4. System strips identifying information. 5. Concern forwarded to counselor queue.
Exception Flow	If message is empty, system shows validation error.



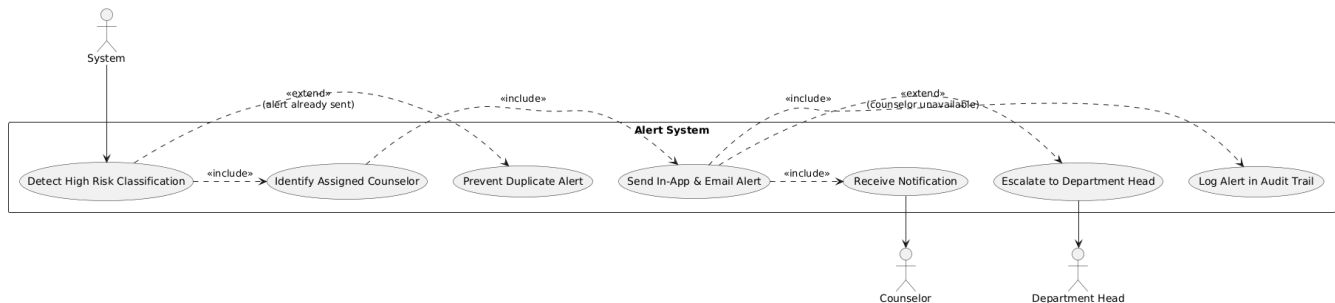
Use Case 6: Track Mood Trends

Actor	Student
Precondition	Student is logged in and has submitted at least 3 daily check-ins.
Postcondition	Mood trend chart displayed with weekly summary.
Main Flow	1. Student opens Mood Tracker. 2. Student logs today's mood (1–5 scale). 3. System stores entry. 4. System renders mood trend line chart.
Exception Flow	If daily check-in already submitted, system shows 'Already checked in today' message.



Use Case 7: Send Counselor Alert

Actor	System (automatic trigger)
Precondition	Student risk level classified as High.
Postcondition	Counselor receives alert notification with student profile link.
Main Flow	1. System detects High risk classification. 2. System identifies assigned counselor. 3. System sends in-app and email alert. 4. Counselor receives notification. 5. Alert logged in audit trail.
Exception Flow	If counselor is unavailable, alert is escalated to department head.



Use Case 8: Send Wellness Reminder

Actor	Notification Service
Precondition	Student has notifications enabled.
Postcondition	Student receives daily wellness reminder on device.
Main Flow	1. Notification service triggers scheduled daily job. 2. System prepares personalized reminder message. 3. System sends push notification to student device. 4. Student receives reminder.
Exception Flow	If delivery fails, system retries up to 3 times before logging failure.

